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Project 1: Theoretically Speaking - Draft 2

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Un Cerebro

Coming from an underserved and under-resourced community, I saw things that most students my age haven't experienced or witnessed. From elementary through middle school, I was pulled out of class to help my mom communicate with her doctors. At the time, I didn't think much about these visits, not until I got older and realized they were mental health related. I remember sitting in a beige waiting room, staring at a painting shaped like a brain. When I was nine, I asked my mom what it was. She said "un cerebro," a brain. Years passed, and I saw how after each year it became more detailed. Its swirls and long swoops reminded me of tree roots spreading across the canvas painting. Those roots mirror something I've learned through my education: mental health doesn't exist in isolation, it spreads through families, communities, and entire systems. This way of thinking—examining systems, integrating multiple perspectives, and making invisible patterns visible—comes directly from the methods of theorists who revolutionized how we study and treat mental health.

Moving to Boston, Massachusetts from Cicero, Illinois was a huge culture and education shock. When I started college at Northeastern, I saw how experiential learning shaped education here. Classes and professors have taught me to think and work in specific ways that I'm only now recognizing as deeply influenced by major theorists in psychology and data science.

My data science class with Dr. John Rachlin showed me the importance of data analysis in medicine, but what I didn't realize was that our methods reflected the work of Urie Bronfenbrenner, who created ecological systems theory. Bronfenbrenner (1979) argued that you can't understand a person in isolation. You must examine all the systems they exist within: their

family, community, and socioeconomic status. His method was to systematically collect data across all these levels and look for patterns.

In my data science project, my teammates and I assessed the effectiveness of screening tools in detecting at-risk mental health patients using The Mental Health Risk Screen (HMHRS) (Carlson, 2024). Here's where I see Bronfenbrenner's method in how Dr. Rachlin taught us: we weren't just looking at individual mental health scores. We examined socioeconomic backgrounds, racial and ethnic identities, and gender representation—all the contextual systems surrounding each individual. Our analysis revealed that individuals from higher socioeconomic backgrounds generally do not have high mental health scores, while those from lower socioeconomic backgrounds tend to report lower self-esteem, higher stress, and more mental health symptoms.

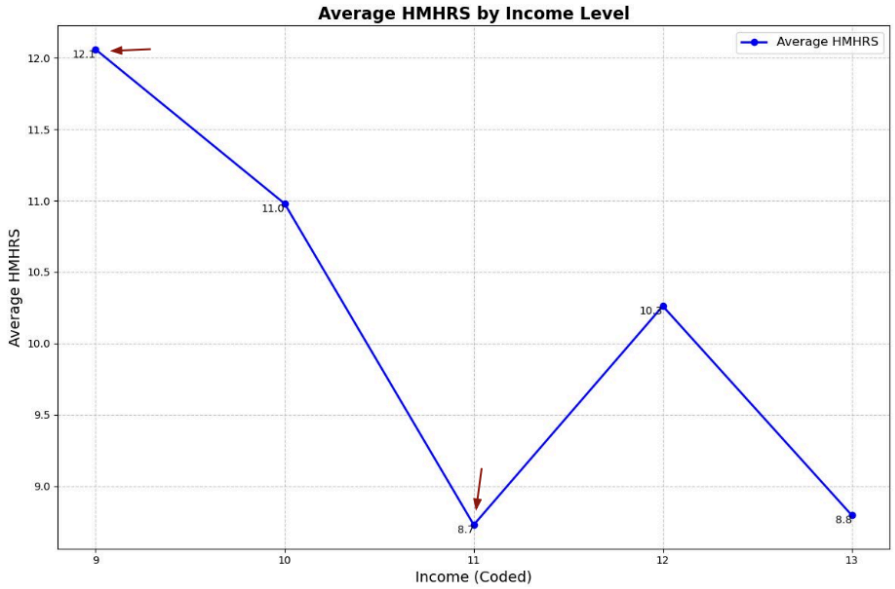

Figure 1. Average HMHRS scores by income level demonstrates a non-linear relationship, with the lowest income group (<$25,000) scoring highest (12.1) and middle-income earners ($50,000 - $74,999) scoring lowest (8.7). Data from Carlson (2024). Created using Python on Spyder for DS2500.

My group and I cross-referenced our findings with existing research showing how families lacking financial resources are less likely to have their basic needs met (Allen et al., 2014). We also found that more gender representation is needed within racial groups, particularly for those identifying as Multiracial/AI/AN. The National Alliance on Mental Illness (2024)

explains how most people of color do not receive the same level of access to care or quality of care compared to white people, contributing to less data representation across mental health screenings.

The assignment structure itself—analyze individual data, zoom out to community patterns, then connect to systemic inequalities—is what and how Bronfenbrenner thought of things. His ecological method taught researchers to always ask: what systems is this person embedded in? Dr. Rachlin's project design forced us to work this way, to never look at a data point without considering its context. Dr. Rachlin had embedded Bronfenbrenner's method so naturally into the assignment that I didn't recognize it as theory—I just thought it was good research practice.

My Clinical Psychology and Mental Health class with Professor Dr. Daniels trained us in the method pioneered by George Engel, a psychiatrist who in the 1970s challenged the purely biological model of medicine. Engel (1977) argued that to truly understand health and illness, practitioners must integrate biological factors, psychological factors, and social/cultural factors together in every clinical encounter.

My first assignment required connecting my lived experiences, cultural knowledge, and class content into one. In class, we wrote our own responses for "What is mental illness?", studied the Diagnostic and Statistical Manual of Disorders (DSM-5), and discussed how the Community Mental Health Act established support networks. This assignment structure directly reflects Engel's biopsychosocial method: we started with our personal/cultural understanding (social), learned the clinical diagnostic framework (biological/psychological), then examined systemic support structures (social again). Dr. Daniels was teaching us to hold all three perspectives simultaneously.

I talked about how, growing up, mental health was never discussed in my family. When I first heard the term, I thought it was related to how you feel just for one day. As I've grown older, I've noticed that machismo (masculine pride) and lack of resources within the Hispanic and Mexican community have influenced false negative narratives for seeking help. Machismo within the Mexican community is shown through "manly" characteristics like being courageous,

showing no emotions, being in control of everything, acting as the head of household, and making all decisions (Zacharias Sexual Abuse Center, n.d.).

Dr. Daniels was pushing us to see through Engel's framework that my mom's mental health couldn't be understood just through her brain chemistry (biological) or even her individual emotions (psychological). I had to consider the cultural narratives about mental health in our community (social), the lack of Spanish-speaking providers (social/systemic), the economic barriers to care (social), and how all of these interact. Engel's method requires this integration, and that's exactly how Dr. Daniels structured our learning—to always push us to consider all three dimensions together.

My co-op as a clinical research coordinator shaped my learning through Northeastern's experiential learning model. Most of my research involved neurofibromatosis and sleep. What caught my eye was how much time we spent creating visualizations: charts, graphs, and sleep pattern diagrams. I've learned this method is seen in the work of John Snow, a 19th-century physician considered a founder of epidemiology. Snow's (1855) famous cholera study didn't just collect data, but he mapped it. His visualization showed where cholera deaths clustered in London, revealing they centered around a specific water pump. The visualization made visible what raw numbers couldn't show.

In my co-op, we created raster plots that showed patients' sleep activity patterns over time. Looking at raw data—numbers showing how many times someone woke up, their pain levels, sleep duration—I could see individual data points, but I couldn't see the pattern. The raster plot changed that. Suddenly, I could see how disrupted sleep correlated with increased pain symptoms the following day, and how this cycle compounded over weeks. The visualization revealed relationships between sleep quality and patients' daily functioning that spreadsheets of numbers had hidden or made confusing. This is Snow's method: making patterns visible so we can understand causes and effects we couldn't see before.

This is the method I was trained to use in my co-op and Dr. Rachlin's class. When we analyzed mental health screening data, we didn't just report percentages. We created visualizations that made disparities visible. Like Snow's map revealing the contaminated pump, our visualizations revealed systemic inequalities affecting mental health access.

That brain painting in the waiting room was my first data visualization. It made visible the challenges inside my mom's head, just as Snow's map made visible the invisible spread of disease, and our visualizations made visible the barriers faced by underserved communities.

I came to Northeastern with curiosity born from those waiting room visits, but I'm leaving with a toolkit of methods refined over decades by theorists who changed their fields. Bronfenbrenner taught me to always look at systems, never individuals in isolation. Engel taught me to integrate biological, psychological, and social factors. Snow taught me that visualization reveals patterns that can change how we understand and address problems.

These methods show up everywhere in my education. The way assignments are structured, the questions professors ask, the research approaches we're taught—they all carry the fingerprints of these theorists. I'm still the same person who sat in that beige waiting room, but now I'm equipped with methods that help me see more clearly, analyze more deeply, and advocate more effectively for communities like the one I came from. When I think about my mom in those waiting rooms, I understand her situation differently now. Not just as an individual with a diagnosis, but as a person embedded in multiple systems—linguistic, economic, cultural—all shaping her access to care. That is what Bronfenbrenner shaped. Her condition can't be reduced to just neurology or psychology or social factors. It's all three interacting. That is what Engel shaped. And the brain painting that fascinated me was teaching me that making the invisible visible is how we understand complex problems. That is what Snow shaped.

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